

Quantitative Metrics for Decomposing Timing and Magnitude Differences in Influenza Forecasts

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Why Does Sub-State Surveillance Matter?

- Public health decisions are often made at the **state level**, but influenza dynamics vary dramatically within states.
- Understanding how local epidemic curves *deviate* from the state curve could improve:
 - Surge capacity planning
 - Resource allocation timing
 - Sub-state early warning

Research Question

Which curve similarity metric best captures the *timing* and *magnitude* differences between HSA-level and state-level influenza incidence?

Key Comparison

DTW and **Fréchet distance** vs. conventional RMSE

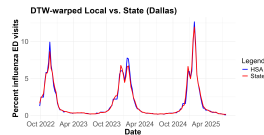
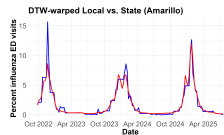
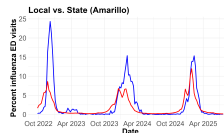
Data & Study Design

Seasons Analyzed

- 2022/23 · 2023/24 · 2024/25
- ~148 total epidemic weeks

Geographic Unit

- ~148 HSAs across the lower 48 states
- Each HSA paired with its parent state curve



Case Studies

Amarillo, TX: high divergence from Texas' state curve

Dallas, TX: low divergence from Texas' state curve

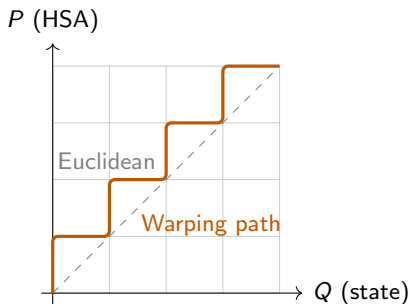
What Is Dynamic Time Warping (DTW)?

Intuition: DTW finds the **optimal elastic alignment** between two time series by allowing non-linear warping of the time axis.

Given sequences $P = (p_1, \dots, p_m)$ and $Q = (q_1, \dots, q_n)$, construct a cost matrix $D \in \mathbb{R}^{m \times n}$:

$$D(i, j) = d(p_i, q_j) + \min \begin{cases} D(i-1, j) \\ D(i, j-1) \\ D(i-1, j-1) \end{cases}$$

$$\text{DTW}(P, Q) = D(m, n)$$



Algorithmic Properties

- Solved via dynamic programming in $O(mn)$
- Boundary conditions: path starts at $(1, 1)$, ends at (m, n)
- Monotonicity: indices never decrease along the path
- Continuity: no skipped indices

Strengths

- Handles phase shifts naturally
- Works on series of unequal length
- Well-studied in speech, ECG, finance

Limitations for Epidemic Curves

- Minimizes *total* cost — can dilate/compress the curve aggressively
- Hard to disentangle timing vs. magnitude
- Not a true metric (no triangle inequality)

What Is Fréchet Distance?

Intuition: Visualize walking your dog on a leash and while you walk along curve P , the dog walks along Q . Fréchet distance is the **minimum leash length** needed over all valid traversals.

Formally, for curves $P, Q : [0, 1] \rightarrow \mathbb{R}^d$:

$$d_F(P, Q) = \inf_{\alpha, \beta} \max_{t \in [0, 1]} d(P(\alpha(t)), Q(\beta(t)))$$

where $\alpha, \beta : [0, 1] \rightarrow [0, 1]$ are continuous, non-decreasing reparametrizations with $\alpha(0) = \beta(0) = 0$, $\alpha(1) = \beta(1) = 1$.

Discrete Version (our implementation):

$$d_F(P, Q) = \min_C \max_{(i, j) \in C} d(p_i, q_j)$$

where C is a valid monotone coupling path through the $m \times n$ grid.

Critical Distinction from DTW

- Fréchet minimizes the **maximum** paired distance
- DTW minimizes the **total sum**
- Fréchet is sensitive to the *worst-case* local mismatch

DTW vs. Fréchet: Side-by-Side

Property	DTW	Fréchet Distance
Optimization target	Minimize <i>sum</i> of distances	Minimize <i>maximum</i> distance
Sensitivity	Global shape similarity	Worst-case local mismatch
Algorithm	DP, $O(mn)$	DP, $O(mn)$
Interpretability	Accumulated mismatch	Bottleneck mismatch
Phase shift handling	Excellent	Good
Magnitude sensitivity	Moderate	High

Why Both?

Each method captures a different facet of curve dissimilarity!

Applying DTW & Fréchet to HSA–State Pairs

Setup

- 1 For each HSA h and its parent state s :
 - Extract weekly incidence vectors $\mathbf{p}_h$ (HSA) and $\mathbf{q}_s$ (state)
 - Align on the same 148-week grid
- 2 Compute $\text{DTW}(\mathbf{p}_h, \mathbf{q}_s)$ and $d_F(\mathbf{p}_h, \mathbf{q}_s)$ within each seasonal peak window
- 3 Extract derived metrics (timing, magnitude, composite)

Timing Metrics

- `signed_lag`: $\bar{i} - \bar{j}$ where i indexes HSA and j indexes state along the warping path
 - Positive $\Rightarrow$ HSA peaks *later*
 - Negative $\Rightarrow$ HSA peaks *earlier*
- `centroid_time_diff`: `hsa_centroid` – `state_centroid` (mass-based timing)

Magnitude Metrics

- `weekly_mag_ratio`: `inc_hsa/inc_state` (week-by-week)
 - $> 1 \Rightarrow$ HSA is consistently *larger*
 - $< 1 \Rightarrow$ HSA is consistently *lower*
- `truth_mag_ratio`: ground-truth peak magnitude ratio

Validating Against Ground Truth

Truth Metrics (from raw ED data)

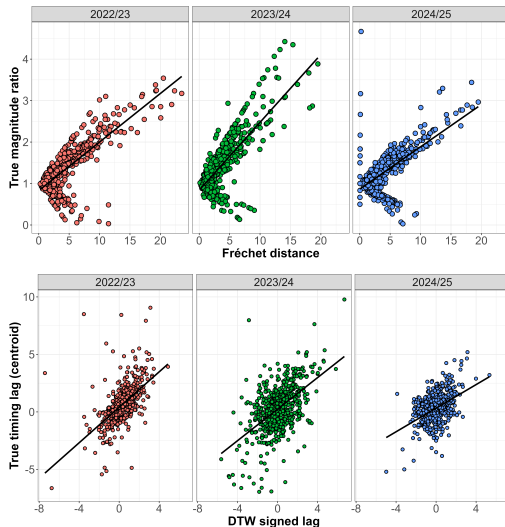
- `truth_time_diff`: actual week-of-peak difference between HSA and state
- `truth_mag_ratio`: ratio of actual peak incidence rates

Validation Strategy

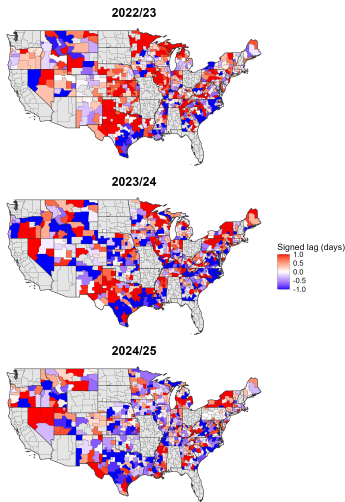
- 1 Compute correlation between metric vs. truth by peak window and season

Strong Correlations

- `signed_lag` $\sim$ `truth_time_diff`
- `frechet` $\sim$ `truth_mag_ratio`



US Maps (Timing)

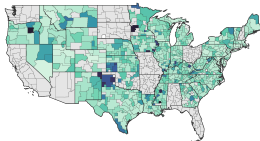


Interpretation

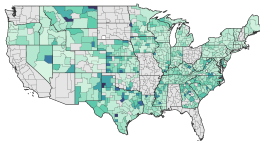
- Warm $\Rightarrow$ HSA peaks *after* state (lagging)
- Cool $\Rightarrow$ HSA peaks *before* state (leading)
- Geographic clusters emerge: Mountain West HSAs often lead; Midwest often lags
- Consistent spatial pattern across seasons suggests structural, not stochastic, divergence

US Maps (Magnitude)

2022/23

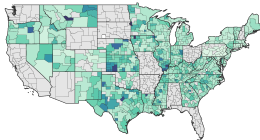


2023/24



Fréchet distance
25
20
15
10
5
0

2024/25



Interpretation

- High frechet (darker) = worst-case local mismatch is large
- Certain areas show higher consistent Fréchet areas, like the top of Texas and Oklahoma region
- Spatial clustering hints at regional flu spread dynamics

Truth Metrics Are Simpler & Interpretable

- `truth_time_diff` is just the plain difference in week-of-peak
- `truth_mag_ratio` is the ratio of peak incidence rates is directly meaningful to epidemiologists
- These are the *ground truth* we validate against

So Why Bother?

- ➊ Truth metrics only capture the **peak** while DTW/Fréchet capture the **full curve**
- ➋ Multi-modal seasons: some HSAs have double peaks where “truth” timing is ambiguous
- ➌ DTW/Fréchet generalize across curve shapes naturally

Limitations

DTW Limitations

- Can over-warp (pathological alignments at endpoints)
- Hard to understand intuitively at times
- Only considers one mapping possibility (deterministic)

Fréchet Limitations

- Sensitive to outlier spikes
- Assumes monotone traversal
- May confound timing and magnitude in a single number

Takeaway

Neither metric is strictly superior to truth metrics. They are *complementary* and most useful when ground truth peaks are unavailable or ambiguous, and most interpretable when validated against truth.

Current State

- Point estimates of DTW/Fréchet per HSA-season-window
- No uncertainty quantification on metric values

Why Probabilistic?

- ED data has inherent noise
- Seasonal epidemic curves are realizations of a stochastic process
- Decision-makers need *confidence intervals*, not just point estimates

Directions Under Consideration

- 1 **SoftDTW** from the `dtwclust` package along with extra code for additional metrics beyond cost
- 2 **Bootstrapping distributions** by resampling weeks within a season to get CIs on `frechet` since there is no established method for a probabilistic version
- 3 **Composite metric formalization** by establishing weights for timing and magnitude components after extensive research

Composite Metric

Motivation

- Timing and magnitude are distinct epidemiological concepts
- Neither DTW nor Fréchet isolates them cleanly
- A composite could provide a single, interpretable divergence score

Candidate Form

$$\mathcal{C}(h, s) = w_1 \cdot \underbrace{|\text{signed_lag}|}_{\text{timing}} + w_2 \cdot \underbrace{|\log \text{mag_ratio}|}_{\text{magnitude}}$$

where w_1, w_2 could be learned from validation correlations.

Longer-Term Goals

- Extend to COVID-19 and RSV curves
- Identify structural predictors of divergence (urbanicity, geography, population size)

Our Work

- 1 Built an R pipeline for HSA–state epidemic curve comparison across 3 flu seasons
- 2 Implemented and compared DTW and discrete Fréchet distance
- 3 Derived timing and magnitude metrics
- 4 Validated against ground truth peak timing and magnitude ratios
- 5 Produced metric scatter plots and US choropleth maps

Key Findings

- Both metrics correlate with truth
- Geographic patterns in divergence are consistent across seasons
- Rural/small-population HSAs diverge most from state curves

Conclusion

DTW and Fréchet offer interpretable, validated alternatives to RMSE for epidemic curve comparison.